

Biology and Resource Management of Cavala preta, *Decapterus macarellus*, in Cabo Verde Waters

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Abstract

Based on various literatures, and various data collected by INDP and the OFCF Project, re-examined biological information was re-examined, constructed life history for Cavala preta (*Decapterus macarellus*). Landing of Cavala preta was strongly correlated with sea surface water temperature. Prediction of Cavala resource was indicated. Throughout analysis of production model and change of target species, directions of resource management for Cavala preta were discussed.

Introduction

Cavala preta, *Decapterus macarellus*, is the most important fishery resource in Cabo Verde. Previous studies on Cavala preta were done by several researchers (Carvalho, 1993; Jardim, 1995; and Almada, 1997). Since 1984, INDP has collected biological data and statistics. In this study, biological information was re-examined, life history was constructed and directions of fishery management was discussed for Cavala preta resource, based on previous results in mentioned above, INDP data and some results obtained from the OFCF Resource Project (herein referenced as to The Project)

Biological Characteristic of Cavala preta

Sex ratio of Cavala preta from data from stock identification survey in 2003 by the Project showed Female : Male was 497: 398. Sex ratio for female was 55.8%, which indicated female predominate. Sex ratio for female against fork length was shown in Figure 1. This showed that most of larger Cavala preta more than 340mm in Fork length (FL) were male dominant. Longevity can be assumed around this size.

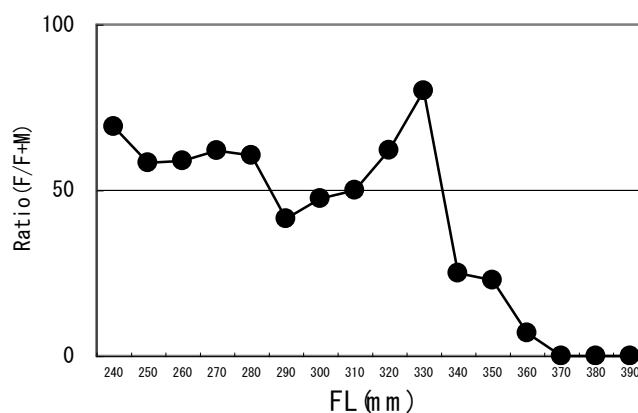


Figure 1. Sex ratio for female of Cavala preta. Data was collected in 2002.

Table 1. Monthly average fork length in cm for female and male, and sex ration in 1998. Data was collected by the INDP.

Month	Female	Male	Sex Ratio
Jan 98	21.6 (35)	21.6 (40)	46.7
Feb	25.0 (28)	25.3 (37)	43.1
Mar	23.8 (44)	24.5 (30)	59.5
April	25.8 (41)	26.3 (34)	54.7
May	22.9 (38)	23.3 (36)	51.4
June	24.9 (57)	25.2 (18)	76.0
July	20.2 (41)	20.4 (30)	57.7
Aug	25.0 (3)	25.3 (75)	3.8
Sep	25.9 (58)	25.5 (17)	77.3
Oct	24.2 (40)	23.8 (35)	53.3
Nov	26.7 (34)	26.5 (41)	45.3
Dec	26.5 (18)	26.2 (57)	24.0

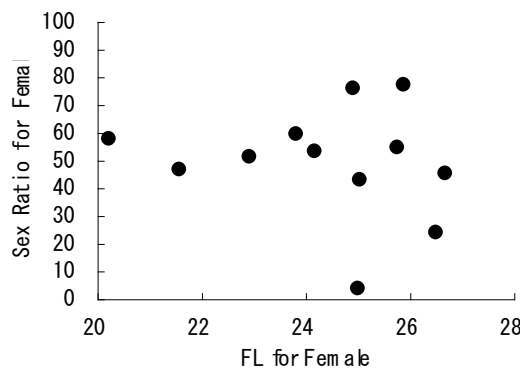


Figure 2. Fork length of female against sex ration, using Table 1.

Table 1 showed monthly average FL for both sex, and sex ration. Data was collected at S. Pedro from S. Vicente Island by the INDP in 1998. Average FL between sexes for each month were quite similar. So, using FL from female and sex ratio in Table 1, a relationship between FL and sex ratio was plotted in Figure 2. From Figure 2, sex ratio became variable when FL was larger than 24cm. Between 24 and 26cm, female was dominated, then male was dominate from larger than 26cm. From Figures 1 and 2, change of growth rate of female and male may start around 25cm in FL. Probably, this is one of causes to bring out variability of sex ratio of Cavala preta.

Spawning Season was examined from change of Gonad Somatic Index (GSI) and larval sampling of the Project. GSI is usually calculated in the following equation:

$$\text{GSI}(\%) = \left(\frac{\text{Gonad weight}}{\text{Body weight}} \right) \times 100.$$

In addition to GSI, change of maturity stage can be insight of spawning season.

Figure 3 showed changes of GSI from April to October 2003. One peak of GSI can be observed. Larvae of Cavala preta were identified (Figure 4), and larval of Cavala appeared during July and August in 2003. These were clearly indicated that spawning season of Cavala was during July and August.

Change of percent of maturity stage of female, which defined in the following:

Stage I	Immature
Stage II	Developing
Stage III	Maturing
Stage IV	Pre-spawning
Stage V	Spawning

Stage VI Recuperating
Stage VII Post-spawning.

Observation of Stage V can provide insight of spawning season (Almada, 1997). According to Jardim (1995), percentage of stage V was high during April to June with sub-peaks in September and November. These discrepancies came from annual changes of spawning or difficulties of clear identification of gonad stages by eyes. Further histological study on gonad is recommended in order to clarify gonad development stages.

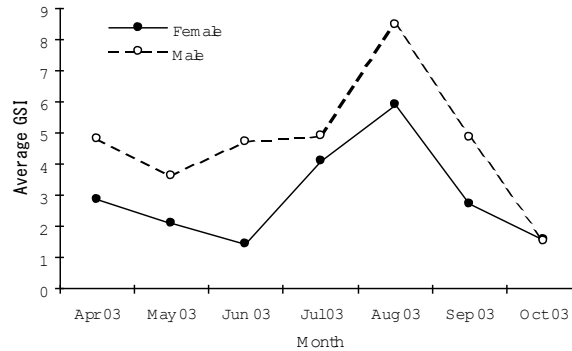


Figure 3. Monthly change of GSI in 2003.

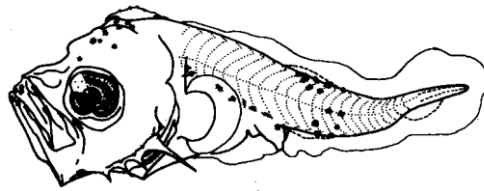


Figure 4. Larvae of Cavala preta, Decapterus macarellus drawn by Kinoshita.

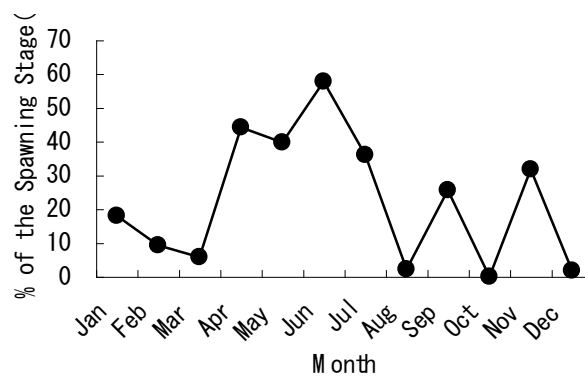


Figure 5. Percentage of stage V from Jardim(1995).

Spawning season of Cavala preta seems to be varied from year to year, but presented data from the Project indicated clear spawning season in July and August.

Figure 6 showed percentage of maturation against FL. Minimum length of maturing was 18 cm in FL. Then maturation rate increased rapidly. Estimated average length of maturation was 20.5cm based on fitting the logistic curve.

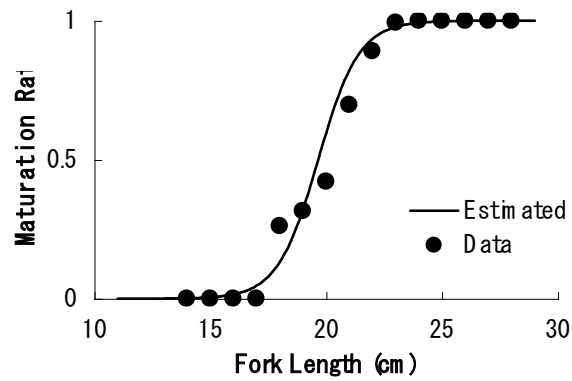


Figure 6. Maturation rate at length. Data was used from Jardim (1995)

Length and weight relationships for male and female were shown in Figure 7, using data collected by the Project from November 2002 to October 2003. An allometric growth model in the following equation was employed:

$$\text{Weight} = a (\text{Length})^b$$

Where a and b are parameters. Parameters were estimated by a linear regression after log transformations for both weight and length. Table 2 showed estimated parameters for female and male. Using Analysis of Covariance (Zar, 1984), comparisons of regression lines did not show no difference between female and male at 5% significant level. Estimated parameters a and b for combined sexes were also showed in Table 2.

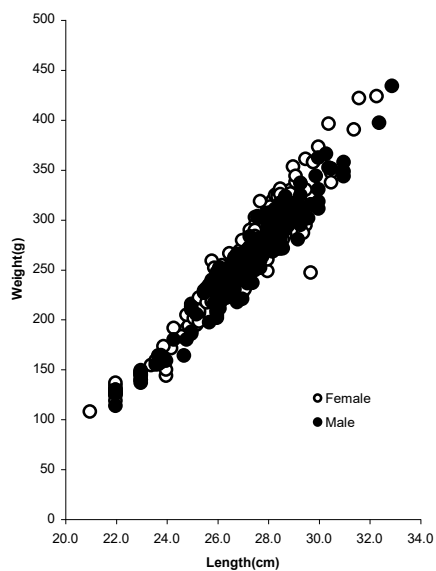


Figure 7. Length-Weight relationship for female and male, using data in Nov 2002 to October 2003 collected by the Project.

Table 2. Estimated parameters of an allometric growth model for female and male, using data in Nov 2002 to October 2003 collected by the Project.

Sex	No Sample	a	b
Female	172	0.00536	3.26857
Male	164	0.00828	3.12461
Combined	336	0.00667	3.19639

The following Table 3 showed previous results of Length and weight relationship. Estimated parameters of the Project data were comparable to previous results.

Table 3. Previous results for length and weight relationships.

Sex	a	b	Author(s)
Female and Male	0.0181	2.87	Carvalho (1987)
Female and Male	0.0078	3.14	Magnusson and Magnusson (1987)
Female and Male	0,0065	3.17	Jardim (1995)
Female	0.0088	3.11	Almada(1997)
Male	0.0147	2.94	Almada(1997)
Female and Male	0.0070	3.17	Almada(1997)

Monthly length frequency of Cavala preta at S. Pedro from 1998 to 1999, using the INDP data was showed in Table 4. The larger Cavala preta (more than 24 or 25cm) were caught from January or February to September. Smaller Cavala preta (less than 20 or 21cm) was captured from October to December or January. In 1998, smaller and larger were captured, but only smaller Cavala preta was caught in December 1999. Between October and January, smaller Cavala preta in size range between 14 to 20or 21cm entered into fishing ground. When smaller Cavala preta entered into fishing ground, larger Cavala preta escaped from the fishing ground. Observation of R/V Sinagoga fishing activities during the Project showed a similar pattern. Length frequency in 1998 unlikely showed this pattern. Although such annual variability may be occurred, switching from small to larger sized Cavala during October to January, then escapement of smaller Cavala preta after January was occurred.

It is not yet clear for this seasonal changes of sized Cavala preta into fishing ground, based on available data.

Assuming spawning in July and recruitment at length 18cm in December, age of recruitment was calculated from previous growth curves. Table 5 showed these results.

Table 5. Estimated age of recruitment.

Author(s)	Growth Parameters			Estimated Age (year)
	L_{∞}	K	t_0	
Carvalho(1987)	34.0	0.49	0	1.5
Jardim (1995)	31.5	0.43	-1.56	0.4
Almada(1997)	30.1	0.34	0.1	2.8

From these results, age of recruitment into fishing ground was varied from 0.4, to 2.8 years, based on the assumption above. These growth parameters were estimated solely from length frequency analysis.

Table 4. Monthly length frequency of Cavala preta at S Pedro, S. Vicente Island from 1998 to 1999, using data from the INDP.

	1998												1999											
FL(cm)	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12
13																								
14										1	1													
15																								
16										35	2													
17										105	18													4
18	1									24	56													72
19	6						12			3	62	20					1							49
20	33	2					33			2	25	53										1	22	
21	38	3	2				27				1	32										2	15	
22	66	5	6	2	17		3		3	5	4	7										2	2	
23	20	12	82	60	41	45	14		11	36	44				1		1					1	1	
24	5	27	162	120	12	116	72		62	58	81	13		2	1		10		2			1		
25	2	56	93	91	5	101	49		45	62	25	105	5	21	8	4	66		13	4			5	
26		40	51	128		53	35		30	14	74	140	67	67	45	22	72	4	48	17			35	
27		15	24	77		18	8		9	1	67	30	70	41	109	73	76	20	82	30			70	
28		6	10	13		4	2		2		9	9	14	3	108	120	62	87	63	15			41	
29			3						1		1	1		1	45	76	31	38	74				24	
30						1									3	11	10	8	35	2			1	
31															3			1	5					

Construction of Life History

The Project has been studying larvae of *Cavala preta*. Larvae of *Cavala preta* from S. Vicente Straight were identified. It can be inferred that spawning ground is around or near coast from this result from the larval study. No juvenile of *Cavala preta* was not collected at the sandy beach survey by a surf-net. Juveniles of *Cavala preta* has been captured by commercial beach seine. A purse seine has captured from immature to adult *Cavala preta*.

During March to September, *Cavala preta* spawn near coastal waters. Juvenile stage stays around coast. *Cavala preta* enters into fishing area as spawning stock when age of 0.5 to 2.5 years is reached.

As showing life history for *Cavala preta* in Figure 8, large number of *Cavala* larvae and eggs were collected in July 2003 by larval net survey. These results suggested spawning period of *Cavala preta* around S. Vicente Island was between July and August. Size of spawners is larger than 20cm in FL.

Results from surf-net sampling did not show appearance of juvenile *Cavala preta* around sandy beach (Table 6). This implied that juvenile of *Cavala preta* would use more off shore around island as nursery area.

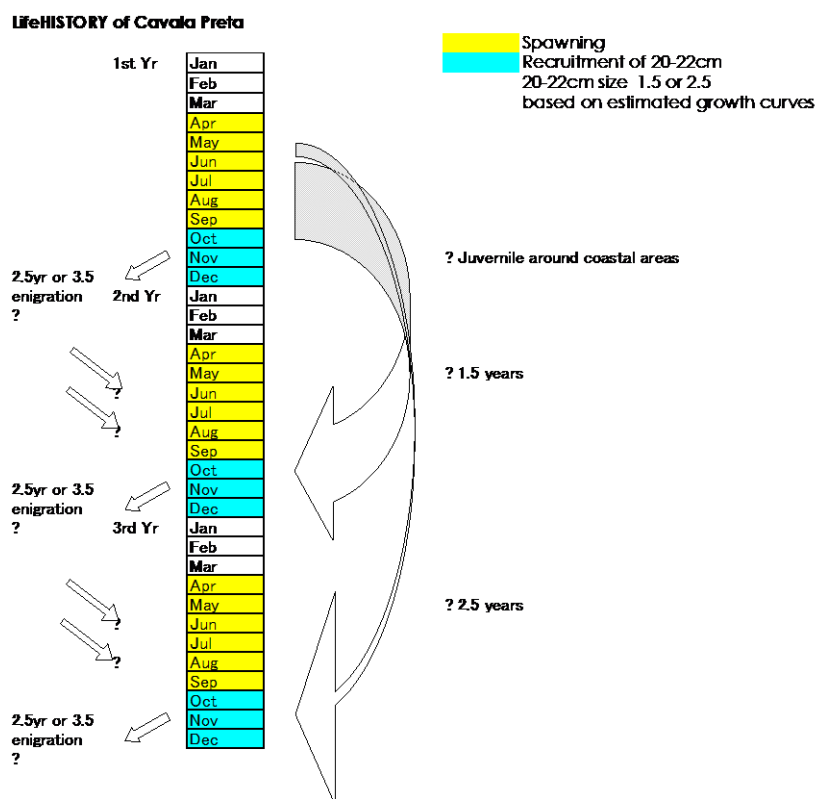


Figure 8. A life history diagram for *Cavala preta*.

According to results of fishing operation survey by R/V Sinagoga, small sized *Cavala preta* (17-22cm) was captured in the fishing grounds during October and December. When these small *Cavala preta* entered to the fishing grounds, medium and large ones (more than 25cm) disappeared. When small ones disappeared from the end of December to the beginning of January, medium and large ones returned to the fishing grounds. This change has not yet fully investigated.

Table 6. Juvenile fish samples in sandy beach in Mindelo from October 2002 to September 2003, using surf-net.

Year	2002												2003												Total
Identified Sp. Gen, Fami	Out 25	Nov 28	Dez 16	Jan 31	Fev 27	Mar 21	Abr 24	Mai 27	Jun 30	Jul 24	Aug 19	Sep 16													
Sardinella sp	3				44				40	10	10		68	175											
Clupeidae				9										9											
Atherina lopeziana	1,138	113	3	2,397	1,627				2,097	4,833	2,722	2,241	9,538	26,709											
Dactylopteryidae						2								2											
Caranx sp		1												1											
Seriola?													1												
Eucinostomus melanopterus			69	90	2	3	1			1				166											
Haemulidae	1		8	65	2	1							2	79											
Diplodus sargus lineatus	3		1	60	2	203	9			40			3	321											
Diplodus fasciatus						1								1											
Oblada melanura?				1										1											
Lithognathus mormyrus													1												
Liza sp.	6	105	2	25	71		1,892	183				5	18	2,307											
Bathygobius casamancus?										1				1											
Scombridae				1										1											
Total Number of Species	5	3	5	8	6	5	3	3	5	2															
Total Number of Fish	1,151	219	83	2,648	1,748	210	1,902	2,320	4,885	2,732	2,246	9,631	29,775												
Total Number of Tows	3	3	3	3	3	2	3	3	3	2	3	3													

Number of captured *Cavala* (larger than 30cm) decreased, and sex ratio become male dominant for larger than 30-33 in Figure 5. After these sizes, population could be naturally decayed, thus it can be considered as terminal size.

Examination of Fishery Management

In Figure 9, recent decrease of landing was observed. This decrease can be speculated as over-fishing or change of oceanographic conditions.

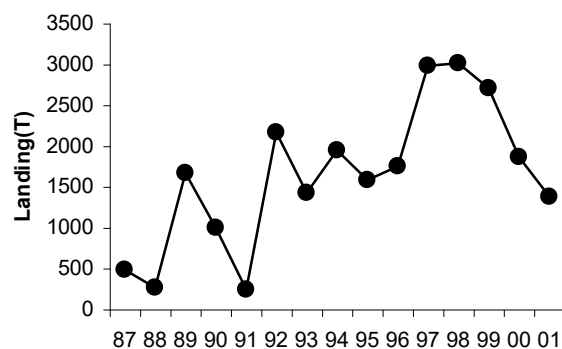


Figure 9. Landing of *Cavala preta* from 1987 to 2001.

First, oceanographic condition was examined, using surface sea water temperature (SST) and landing. In Cabo Verde, landing can be considered as index of resource condition because of some constant of number of boats operated. Spawning and consequent recruitment could be influenced by water temperature. Average SST between July and August (Cavala spawning period) and landings were plotted from 1989 to 2003. Two years lag was found in observing this relationship, so SST was shifted two years (Figure 10). Landings were positive relationship with SST of two-years lag.

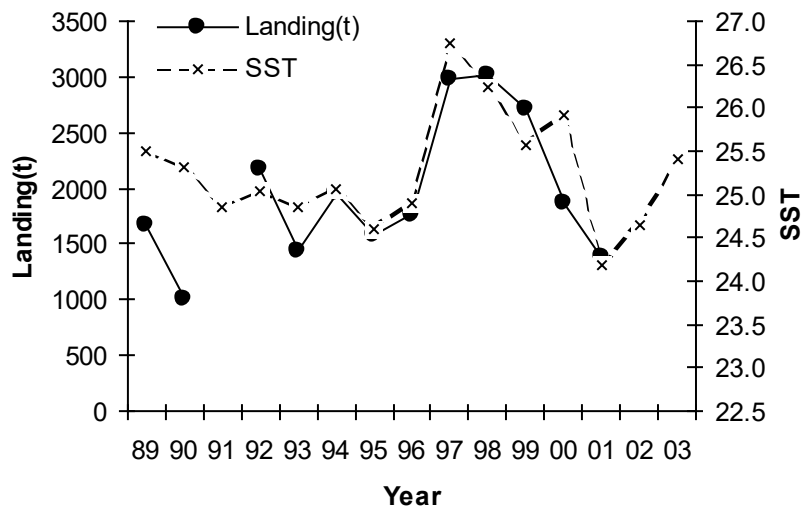


Figure 10. Relationship between SST of two-years lag and landings of Cavala preta.

Relationship between SST of two-years lag and landings was:

$$(\text{Landing})_{t+2} = -1.47 + 657 \times (\text{SST})_t \quad (r=0.729)$$

where t is year.

Based on SSTs in 2000 and 2001, estimated landings in 2002 and 2003 were 1,500 and 2,000 tons, respectively.

This two-years lag could be explained as recruitment in Figure 8. Age of recruitment was 2 year, and this 2-year group could be considered as main landing of Cavala preta.

This result only indicated this relationship between SST of two-years lag and landings. Some explicable reasons were made, but there were no data to explain this relationship. Larval survey and oceanographic monitoring give more plausible causes and prediction of resource can be possible.

Effects of fishing pressure for two stocks in northern and southern waters were examined, based on results of stock structure. Using Schaefer Surplus Production Model (1954), effects of fishing were examined for both areas. Results were shown in Figure 11. Estimated Maximum Sustainable Yield (MSY) and efforts at MSY for both waters were shown in Table 7.

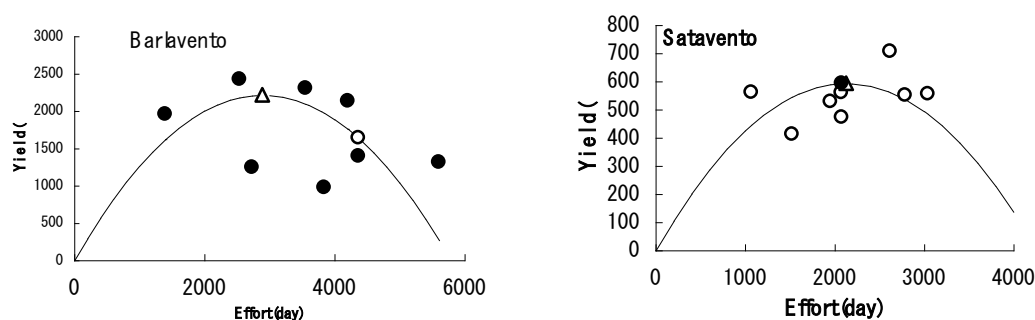


Figure 11. Results of fitting for Surplus Production Model for northern and southern stocks of Cavala preta.

Table 7. Estimated MSY and effort at MSY for northern and southern stocks of Cavala preta.

Waters	MSY(t)	Effort (Days) at MSY
Northern	2213	2893
Southern	593	2130

Historically, fishing efforts surpassed at those of MSY levels. Resource of Cavala preta was influenced by oceanographic condition in the above mentioned, however, resource has a certain carrying capacity. So, Table 7 is some indication of criteria to manage Cavala preta resource. Furthermore, Figure 12 showed relationship between Cavala preta and Olho largo, *Selar crumenophthalmus*. When Cavala preta decreased, Olho largo increased. This mutual shifting emphasize that we have to observe not only Cavala preta but also Olho largo.

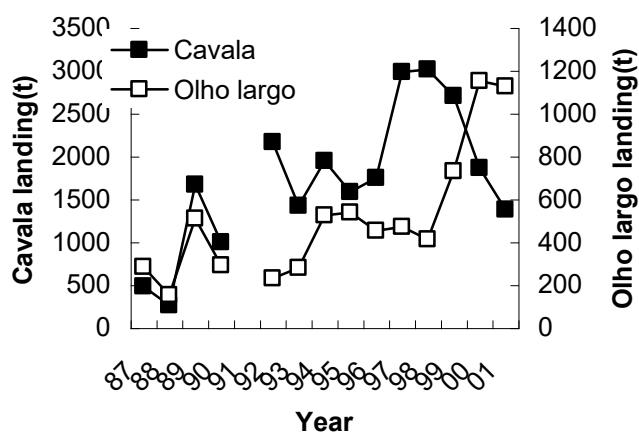


Figure 12. Relationship of landings between Cavala preta and Olho largo, *Selar crumenophthalmus*.

Although Cavala preta resource can be affected by environmental condition, some limitation can be existed. In addition to these, utilization of resource can be swiftly changed from one to another species. These factors could be considered as some directions of resource management for Cavala preta,

Literatures

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